

Comparative Evaluation of a Grounded Singapore Tourism Agent

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Executive Summary

This report evaluates a retrieval-augmented Singapore tourism agent using two generation models: openai/gpt-4o-mini and openai/gpt-6.1-sol-pro, both accessed through OpenRouter. The comparison uses the same 30-question test set, corpus snapshot, retrieval settings and four-agent-tool design. GPT-6.1 selected the expected tools much more consistently, reaching 90.0% automatic tool-selection accuracy compared with 46.7% for GPT-4o-mini. Human review of ten common questions also favoured GPT-6.1, with an 80.0% overall pass rate compared with 30.0%. However, automatic Judge scores did not consistently agree: GPT-6.1 received only 23.3% automated faithfulness despite 80.0% human faithfulness, while its deterministic validator pass rate fell to 13.3%. The results indicate a substantial improvement in practical answer quality and orchestration, but also expose weaknesses in the evaluation design. A fixed independent judge and stricter evidence-level validation are needed before drawing a final model-selection conclusion.

1 Project Objective

The project addresses a common tourism information problem: visitors need accurate answers about opening hours, prices, closures, rules, accessibility and transport, but these facts are distributed across many official websites and often change by date. The goal was to build an assistant that retrieves official evidence, separates frozen snapshot information from current checks, cites the source used and abstains when a claim cannot be verified. The system is an advisory prototype rather than a booking service or real-time guarantee.

2 System and Data

The prototype was implemented in Python in Google Colab. Local retrieval uses sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2 embeddings with a FAISS index, top_k of 6 and a minimum similarity threshold of 0.25. Four tools are available: search_knowledge_base, get_route_information, fetch_live_official_page and search_live_official_web. The official Singapore tourism snapshot is dated 4 October 2026. Its manifest records 67 attempted documents, of which 35 were usable, 30 failed extraction and two required manual review. The 35 usable documents contain 66,605 text characters divided into 57 searchable chunks. Dynamic pages from organisations such as Sentosa, Visit Singapore and the National Heritage Board produced major coverage gaps.

3 Evaluation Design and Comparability

The test set contains 30 manually designed questions: eight time-sensitive questions, six visitor-rule questions, six route or accessibility questions, four multi-source itinerary tasks, three unanswerable questions and three ambiguous or adversarial prompts. The archived question sets are identical across the two model runs. Evaluation combines a deterministic validator, an LLM Judge and manual review of the same ten question IDs: Q01, Q06, Q09, Q13, Q15, Q19, Q21, Q24, Q25 and Q29. GPT-6.1's ten reviewed answers exactly match its archived result rows. The GPT-4o-mini review uses the same questions, but its reviewed answer text is not identical to the final corrected archive, suggesting that it may represent an earlier run. Human comparisons are therefore informative but not a perfectly controlled paired test. In addition, the judging code uses the configured generation model as the Judge. The two automatic Judge result sets consequently do not share a fixed evaluator, limiting direct comparison of Judge-based metrics.

Automatic Metric Comparison

Metric	GPT-4o-mini	GPT-6.1	Change
Tool selection	46.7%	90.0%	+43.3 pp
Validator pass	43.3%	13.3%	-30.0 pp
Judge faithfulness	66.7%	23.3%	-43.3 pp
Snapshot correctness	43.3%	66.7%	+23.3 pp
Citation support	83.3%	60.0%	-23.3 pp
Uncertainty	46.7%	53.3%	+6.7 pp
Constraint satisfaction	66.7%	63.3%	-3.3 pp

4 Automatic Evaluation Results

GPT-6.1 showed a clear orchestration advantage. Corrected tool-selection accuracy increased by 43.3 percentage points, from 46.7% to 90.0%. Snapshot correctness rose from 43.3% to 66.7%, and uncertainty handling improved from 46.7% to 53.3%. Other automatic measures moved in the opposite direction. The deterministic validator fell from 43.3% to 13.3%, automated citation support fell from 83.3% to 60.0%, automated faithfulness fell from 66.7% to 23.3%, and constraint satisfaction decreased slightly from 66.7% to 63.3%. The low validator rate despite strong tool selection suggests that selecting a tool did not guarantee compliant final output, including official URLs, evidence dates and support for every claim. Category results reinforce this distinction. GPT-6.1 achieved 100.0% tool selection on multi-source itinerary, route or accessibility, unanswerable and visitor-rule questions, and 87.5% on time-sensitive questions. However, its time-sensitive validator pass rate was 0.0% and snapshot correctness was only 37.5%. It also reached only 33.3% tool selection on ambiguous or adversarial questions. By comparison, GPT-4o-mini passed the deterministic validator on 75.0% of time-sensitive questions despite much weaker overall tool selection. These patterns suggest that orchestration and final-answer compliance should be reported separately rather than collapsed into one success score. The Judge changes must be interpreted cautiously because the evaluator model changed between runs.

Human Review Comparison

Metric	GPT-4o-mini	GPT-6.1	Change
Tool selection	40.0%	100.0%	+60.0 pp
Faithfulness	30.0%	80.0%	+50.0 pp
Snapshot correctness	33.3% (n=9)	80.0% (n=10)	+46.7 pp
Citation support	20.0%	90.0%	+70.0 pp
Uncertainty	50.0%	90.0%	+40.0 pp
Constraint satisfaction	40.0%	70.0%	+30.0 pp
Overall pass	30.0%	80.0%	+50.0 pp

5 Human Review Results

Manual assessment presents a different picture. GPT-6.1 achieved 100.0% tool-selection correctness, 80.0% faithfulness, 80.0% snapshot correctness, 90.0% citation support, 90.0% uncertainty handling, 70.0% constraint satisfaction and an 80.0% overall pass rate. The corresponding GPT-4o-mini figures were 40.0%, 30.0%, 33.3% across nine applicable questions, 20.0%, 50.0%, 40.0% and 30.0%. The largest apparent improvements were citation support at 70.0 percentage points, tool selection at 60.0 points and overall pass and faithfulness at 50.0 points each. GPT-6.1 failed overall on two reviewed questions. For Q09 it contradicted the evaluation snapshot concerning outside food at Singapore Zoo. For Q19 it omitted several expected wheelchair-accessible Sentosa options and mishandled the evidence for Sentosa Express.

6 Interpretation

The strongest evidence in favour of GPT-6.1 is its tool behaviour and the human assessment of its archived answers. It followed the multi-tool workflow more consistently and produced answers that reviewers regarded as better supported, more appropriately qualified and more complete. This is valuable for an agent whose reliability depends on retrieving the right source before generating an answer. Nevertheless, the comparison also demonstrates that evaluation metrics can fail in different directions. The automatic Judge marked GPT-6.1 faithfulness at 23.3%, while humans assigned 80.0%; faithfulness disagreement on the ten reviewed GPT-6.1 questions was 60.0%. Snapshot, citation, uncertainty and constraint disagreement rates were 20.0%, 20.0%, 20.0% and 10.0%. These gaps show that an LLM Judge should not be treated as ground truth. Citation presence is also insufficient: a response may list an official page while the page does not support every claim. Future validation should operate at claim level and require a supporting evidence span for each factual statement.

7 Limitations and Responsible Use

The corpus remains incomplete because many JavaScript-rendered official pages could not be extracted. The ten-question human sample is small, and the archived GPT-4o-mini review is not text-identical to the final run. One earlier reference answer was unrelated to its question, and another visitor-rule reference conflicted with the cited official page, so test-set quality also affects measured performance. The current experiments do not report average response latency, average API cost or an end-user usability score: [DATA NOT AVAILABLE]. The agent must not purchase tickets, make reservations or guarantee visa, legal, medical or emergency outcomes. Time-sensitive claims should always include a Singapore-date check and a directly supporting official source.

8 Conclusion and Next Steps

On the available evidence, GPT-6.1 is the stronger candidate for the next prototype because it achieved 90.0% automatic tool-selection accuracy and 80.0% human overall pass. This conclusion is provisional rather than final. The next experiment should freeze one independent Judge model, regenerate human-review sheets directly from each archived run, repair defective expected answers and record complete tool traces. Category-level error rates, response latency, token use and cost should also be measured. These changes would convert the current promising comparison into a reproducible model-selection study suitable for a deployment decision.

Evidence Archive

Compared experiment packages: gpt4o_mini_baseline_20261004_122103.zip and gpt6_1_baseline_20261004_144053.zip. Average latency, API cost and end-user usability measurements remain [DATA NOT AVAILABLE].